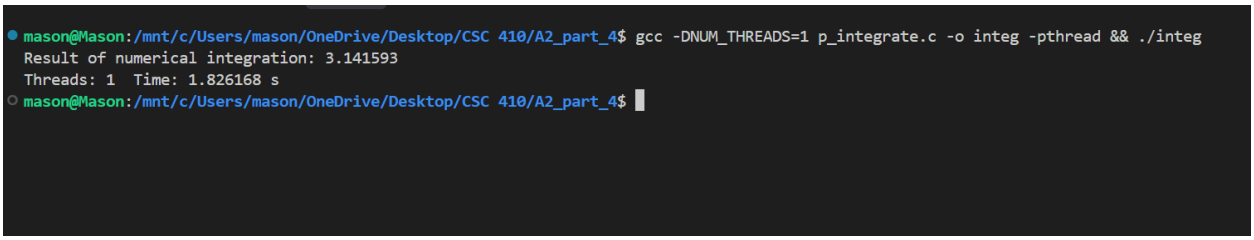
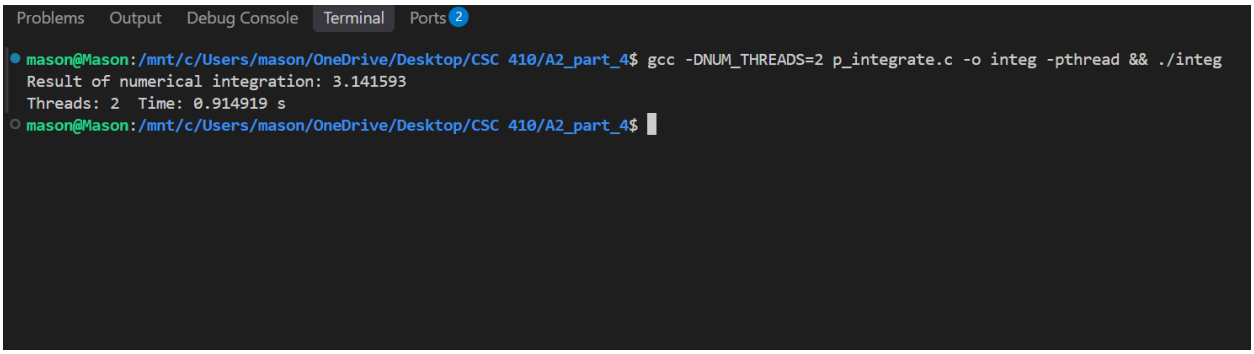
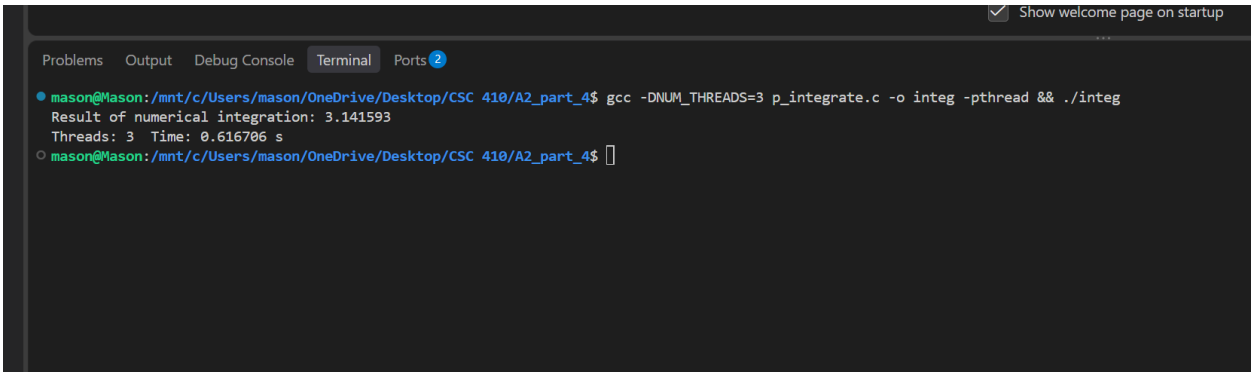
A2 Part 4 - Screenshots

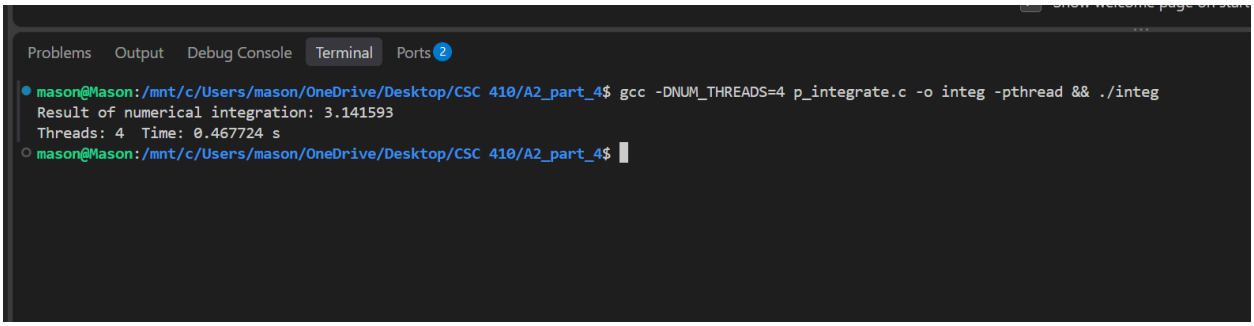
Mason Stanley

Task 1: Numerical Integration (1-4 threads)









Task 2: Merge Sort Approach 1 - Segment-based (1-4 threads)

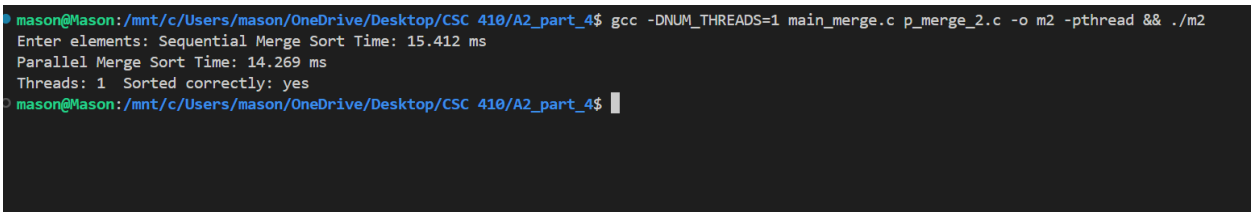
```
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$ gcc -DNUM_THREADS=1 main_merge.c p_merge_1.c -o m1 -pthread && ./m1
Enter elements: Sequential Merge Sort Time: 14.842 ms
Parallel Merge Sort Time: 15.181 ms
Threads: 1  Sorted correctly: yes
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$
```

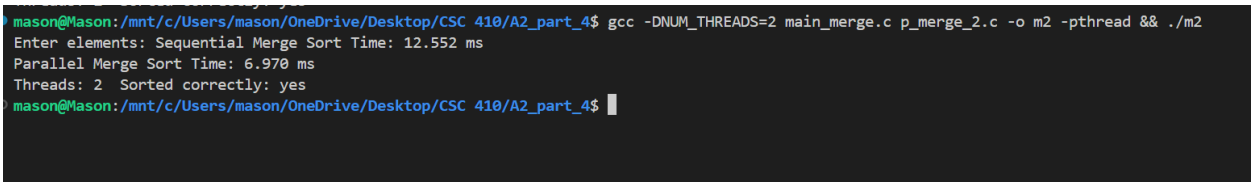
```
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$ gcc -DNUM_THREADS=2 main_merge.c p_merge_1.c -o m1 -pthread && ./m1
Enter elements: Sequential Merge Sort Time: 14.920 ms
Parallel Merge Sort Time: 6.962 ms
Threads: 2  Sorted correctly: yes
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$
```

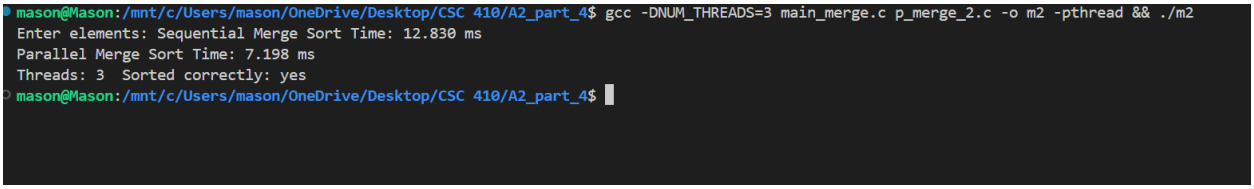
```
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$ gcc -DNUM_THREADS=3 main_merge.c p_merge_1.c -o m1 -pthread && ./m1
Enter elements: Sequential Merge Sort Time: 12.335 ms
Parallel Merge Sort Time: 5.345 ms
Threads: 3  Sorted correctly: yes
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$
```

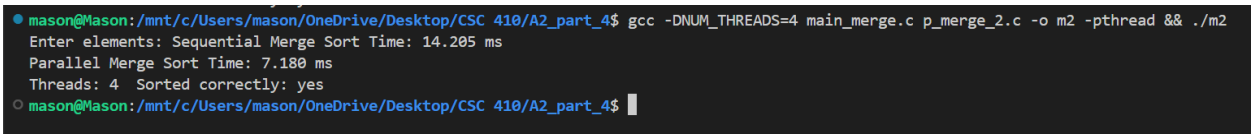
```
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$ gcc -DNUM_THREADS=4 main_merge.c p_merge_1.c -o m1 -pthread && ./m1
Enter elements: Sequential Merge Sort Time: 12.180 ms
Parallel Merge Sort Time: 7.381 ms
Threads: 4  Sorted correctly: yes
mason@Mason:/mnt/c/Users/mason/OneDrive/Desktop/CSC 410/A2_part_4$
```

Task 2: Merge Sort Approach 2 - Recursive thread creation (1-4 threads)









Task 2: Merge Sort Approach 3 - Thread pool (1-4 threads)

